

DATA 642 – Sample Final Exam

Module 1 — Optimization Foundations

1. Explain how gradient descent minimizes a differentiable function and describe how step size affects convergence.
2. Describe the role of the Hessian in determining curvature and explain when Newton's method is advantageous.
3. Explain the geometric meaning of projecting onto a subspace in linear regression.
4. Describe how eigenvalues influence the convergence of optimization algorithms.
5. Explain the relationship between the Gram matrix and pairwise inner products.
6. Describe how the conditioning of a matrix affects optimization in regression problems.
7. Compare gradient descent and coordinate descent in terms of updates and convergence behavior.
8. Explain how adding regularization changes the loss landscape and improves stability.
9. Describe how stochastic gradient descent approximates full gradients.
10. Explain the concept of projection onto convex sets in constrained optimization.

Module 2 — Sparsity-Aware Learning

11. Describe how the LASSO objective leads to sparsity and explain the role of soft thresholding.
12. Explain why ridge regression cannot produce sparse solutions.
13. Compare ridge and LASSO in terms of geometry, optimization, and sparsity.
14. Describe the coordinate descent algorithm for solving LASSO.
15. Explain motivation for sparsity-aware learning in high-dimensional problems.
16. Describe the effect of feature scaling on sparsity-inducing methods.
17. Explain how cross-validation is used for selecting regularization parameters.
18. Describe how elastic net combines properties of ridge and LASSO.
19. Explain practical considerations when running LASSO on correlated data.
20. Describe situations where sparsity-aware learning provides major advantages.

Module 3 — Hilbert Spaces, SVMs, and Kernel Ridge Regression

21. Describe what a Hilbert space is and explain why it is useful in machine learning.
22. Explain the geometric idea behind maximum-margin classification.
23. Describe the primal soft-margin SVM formulation and the meaning of slack variables.
24. Explain how the dual formulation enables the kernel trick.
25. Describe the conditions of Mercer's theorem for valid kernels.
26. Explain how support vectors define the SVM decision boundary.
27. Describe the idea behind reproducing kernel Hilbert spaces (RKHS).
28. Explain kernel ridge regression and describe its closed-form solution.
29. Describe the effect of kernel choice on model complexity and smoothness.
30. Compare hinge loss and logistic loss in classification problems.

Module 4 — PCA, ICA, CCA, IVA, Matrix/Tensor Factorization, Multimodal Learning

31. Describe the full PCA procedure and explain how variance maximization identifies principal components.
32. Explain PCA via SVD and describe why SVD provides a stable computational method.
33. Explain how PCA reconstructs data from low-dimensional representations.
34. Describe whitening and explain its importance in ICA.
35. Explain ICA's goal of maximizing non-Gaussianity to obtain independent components.
36. Describe the role of kurtosis and negentropy in ICA.
37. Explain the FastICA algorithm and why fixed-point iterations converge quickly.
38. Describe key differences between PCA, ICA, CCA, and IVA.
39. Explain the idea of multimodal learning and how feature fusion can be achieved.
40. Describe matrix and tensor factorization and explain how they uncover latent structure.

Module 5 — Clustering, GMMs, Anomaly Detection

41. Describe the k-means algorithm and explain why it decreases the objective each iteration.
42. Explain how k-means++ selects initial centers.
43. Compare centroid-based clustering with density-based clustering.
44. Describe the DBSCAN algorithm and explain core, border, and noise points.
45. Explain how GMMs model multi-cluster structure using Gaussian components.
46. Describe the EM algorithm for GMMs and explain the E-step and M-step.
47. Explain how GMMs can be used for anomaly detection.
48. Describe silhouette width and explain how it measures clustering quality.
49. Explain hierarchical clustering and compare linkage methods.
50. Describe spectral clustering and explain how graph Laplacians are used.

Module 6 — Neural Networks, Autoencoders, VAEs

51. Describe how a feedforward neural network computes nonlinear functions.
52. Explain the backpropagation algorithm and how gradients flow.
53. Describe the vanishing gradient problem and why ReLU-like activations mitigate it.
54. Explain the architecture and training goal of a basic autoencoder.
55. Describe why mean squared error is a common reconstruction loss.
56. Explain denoising autoencoders and why noise improves representation learning.
57. Explain the conceptual structure of a variational autoencoder.
58. Describe the reparameterization trick and why it is needed.
59. Explain batch normalization and how it stabilizes training.
60. Describe dropout and explain how it acts as regularization.